

Course title	Pharmaceutical Organic Chemistry I
Course code	PR 101
Department/s that participated in the teaching	Department of Pharmaceutical Organic Chemistry
Total number of credit hours of the course	3 Credit Hours (4 Contact hours): Lectures: 2 (2) Practical:1 (2)
Course Type	Compulsory
Academic level at which the course is taught	Level I
Semester	Fall
Academic program	Bachelor of Pharmacy (PharmD)
Faculty/University	Faculty of Pharmacy, Modern University for Technology and Information.
Course Coordinator	Dr. Nirvana Ali Gohar
Course Specification Approval	Department council
Date of Approval	September 2025

Number of the Week	Scientific content of the course (Course Topics)
	Theoretical topics
1	Practical Topics
	Introduction, bonding, electronegativity, Hybridization
2	Safety in the organic chemistry laboratory
	Introduction, bonding, electronegativity, Hybridization
3	Chem Sketch program- Solubility scheme
	Alkane and cycloalkanes
4	Identification of Aliphatic Alcohols
	Alkene and alkadienes
5	Identification of Amines
	Alkyne
6	Identification of salts Amines
	Stereochemistry
7	Identification of phenols
	Midterm Exam
8	Stereochemistry
	Identification of aliphatic aldehydes & ketones
9	Stereochemistry
	Identification of aromatic aldehydes & ketones
10	Alkyl halides, Nucleophilic substitution reaction
	Identification of Carboxylic acids
11	Aromaticity
	Identification of salts of Carboxylic acids
12	Electrophilic substitution reaction
	Identification of Hydrocarbons
13	Electrophilic substitution reaction (Orientation)
	Practical Exam I
14	Summary of key concepts and Discussions
	Practical Exam II
Final Written Exam*	

Syllabus of theoretical course for first and second levels (Fall Semester) for the academic year 2025-2026

Course	Pharmaceutical Organic Chemistry I (PR 101- CPR 101(3 Groups))				
Day	Sunday	Wednesday		Thursday	
Time	1:00-3:00	9:00-11:00	11:00 – 1:00	11:00-1:00	1:00-3:00
Place	Theatre (4)			Theatre (1)	
Group	3	5	4	1	2
Week(1) (21-25/9/2025)	Introduction, bonding.... (Dr. Nirvana)	Introduction, bonding.... (Prof. Eman)			
Week(2) (28/9- 2/10/2025)	Introduction, bonding.... (Dr. Nirvana)	Introduction, bonding.... (Prof. Eman)			
Week(3) (5- 9/10/2025)	Alkane and cycloalkanes (Prof. Shaimaa)	Alkane and cycloalkanes (Prof. Asmaa)			
Week(4) (12- 16/10/2025)	Alkene and alkadienes (Prof. Shaimaa)	Alkene and alkadienes (Prof. Asmaa)			
Week(5) (19- 23/10/2025)	Alkyne (Dr. Nirvana)	Alkyne (Prof. Asmaa)			
Week(6) (26- 30/10/2025)	Stereochemistry (Prof. Asmaa)	Stereochemistry (Prof. Eman)			
Week(7) (2- 6/11/2025)	Midterm				

Syllabus of theoretical course for first and second levels (Fall Semester) for the academic year 2025-2026

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Day	Sunday	Wednesday		Thursday	
Time	1:00-3:00	9:00-11:00	11:00 – 1:00	11:00-1:00	1:00-3:00
Place	Theatre (4)			Theatre (1)	
Group	3	5	4	1	2
Week(8) (9- 13/11/2025)	Stereochemistry (Dr. Nirvana)	Stereochemistry (Prof. Eman)			
Week(9) (16- 20/11/2025)	Stereochemistry (Dr. Nirvana)	Stereochemistry (Prof. Eman)			
Week(10) (23- 27/11/2025)	Alkyl Halides (Dr. Nirvana)	Alkyl Halides (Prof. Eman)			
Week(11) (30/11- 4/12/2025)	Aromaticity (Dr. Nirvana)	Aromaticity (Dr. Nirvana)			
Week(12) (7- 11/12/2025)	ES (Prof. Eman)	ES (Prof. Asmaa)			
Week(13) (14- 18/12/2025)	ES (orientation) (Prof. Eman)	ES (orientation) (Prof. Asmaa)			
Week(14) (21- 25/12/2025)	Summary of key concepts (Prof. Eman)	Summary of key concepts (Prof. Asmaa)			

Methods of students' assessment:

NO.	Assessment Methods		Assessment Timing (Week Number)	Marks		Percentage% of total course marks
1.	Periodical work	Exam Semester work (Quiz)	4 th -11 th	10	20	13.3%
2.		Assignment (Project)	3 rd – 11 th	10		
3.	Final Practical Exam		13 th -14 th	40		26.7%
4.	Written exams	Midterm exam	7 th	25	75	50%
5.		Final written exam*	--	50		
6.	Final Oral Exam*		--	--	15	10%

Structure of Organic Compounds

Atomic orbital

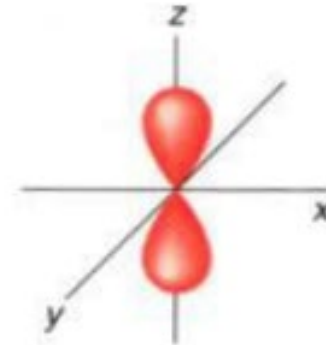
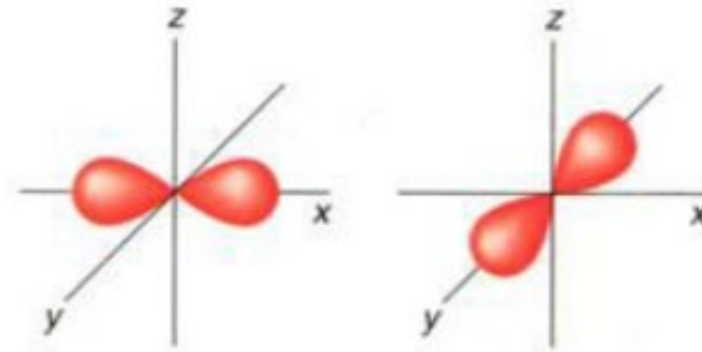
It is the region in the atom to which an electron is confined and which has a geometrical shape. The orbital is occupied by two electrons, one of them spins clockwise and the other spins anticlockwise.

S orbital has **spherical shape** around the nucleus.

P orbitals **have dumb-bell shape** (P_x , P_y , P_z)

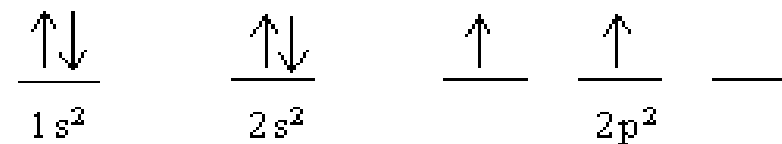


s orbital

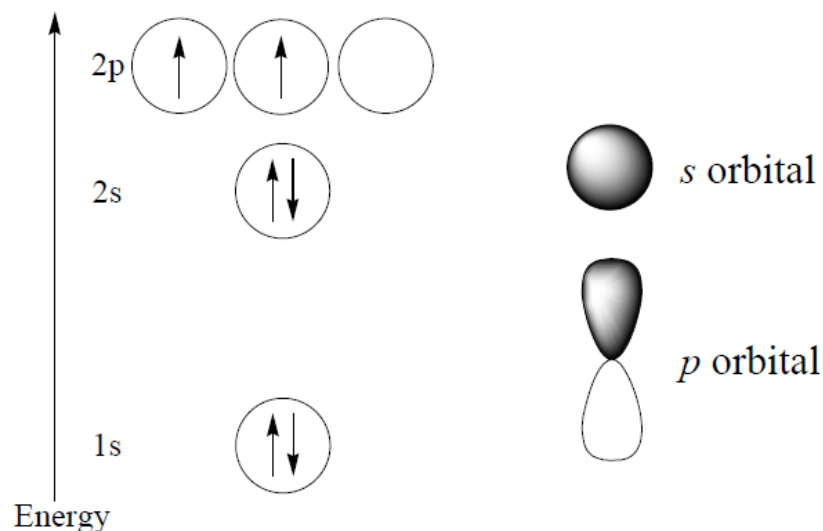


***p* orbitals**

Atomic Orbitals on Carbon



- Carbon atom it has **four valence electrons**. It satisfies the **octet rule** in compounds by **sharing electrons**.

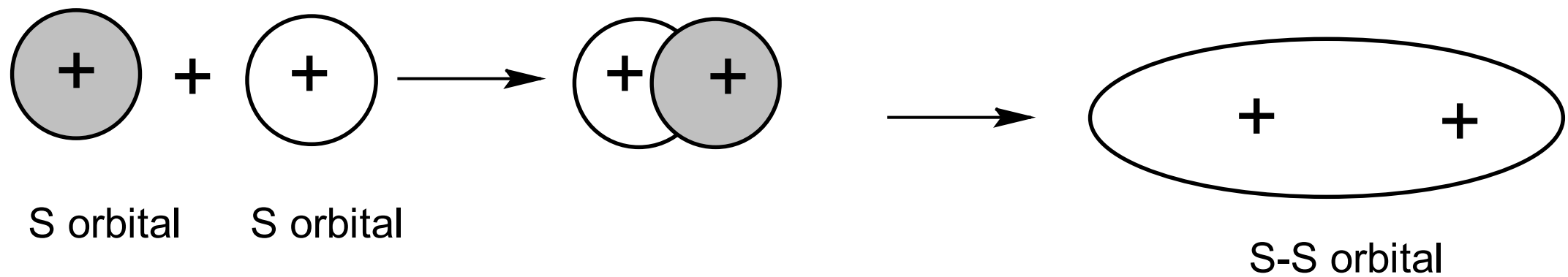


- These are the orbitals that exist on atomic carbon (not connected to anything).

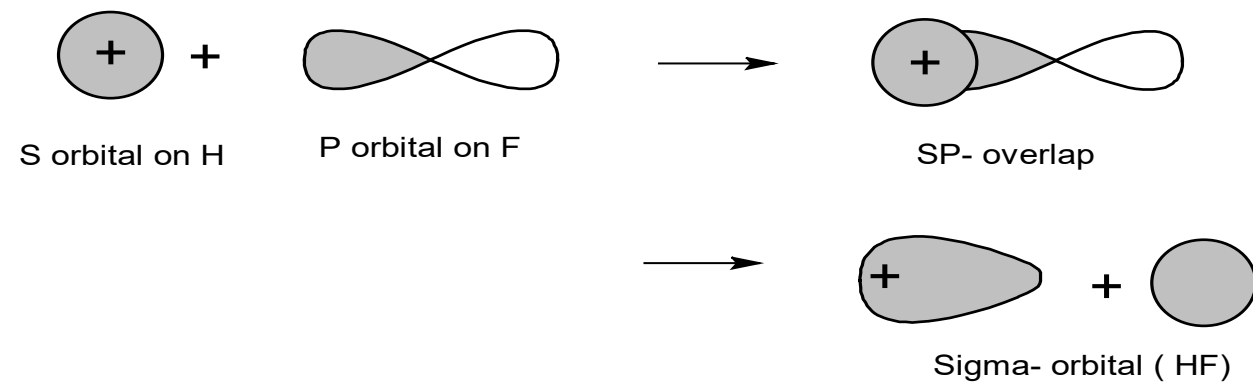
Molecular orbitals (M.O)

The molecular orbital is formed when two atomic orbitals overlap. So that the electrons come to occupy the common orbital which surrounds two nuclei.

Two isolated H atoms



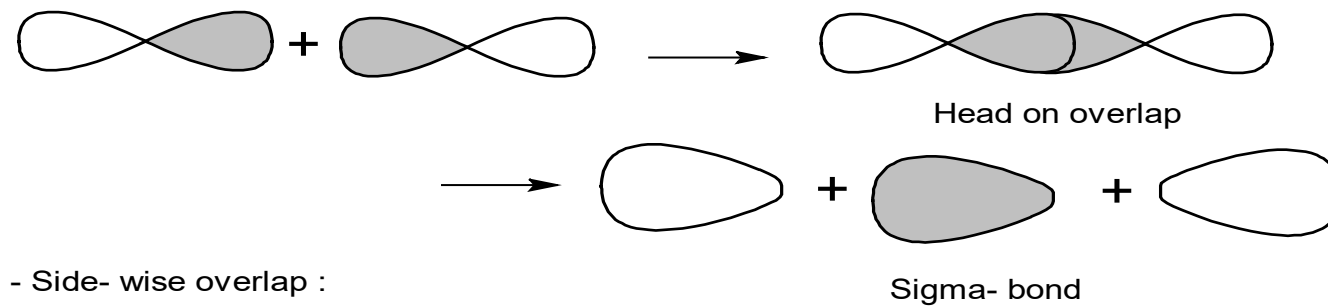
S-P orbitals:



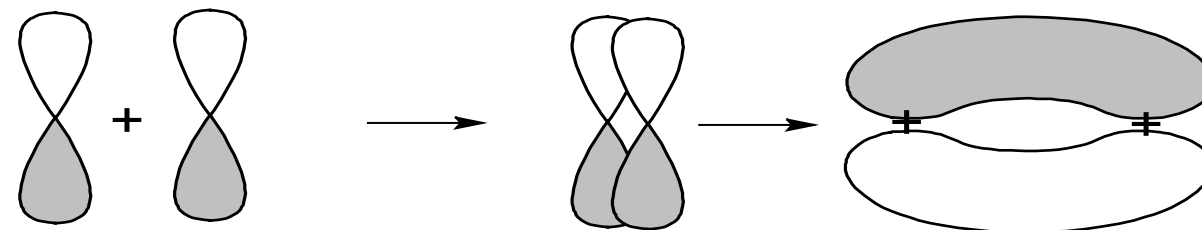
P-P orbitals:

It may take place two ways:

A- Head on overlap: (Sigma strong)



B - Side-wise overlap :



π – bond

π – bond is characterized by:

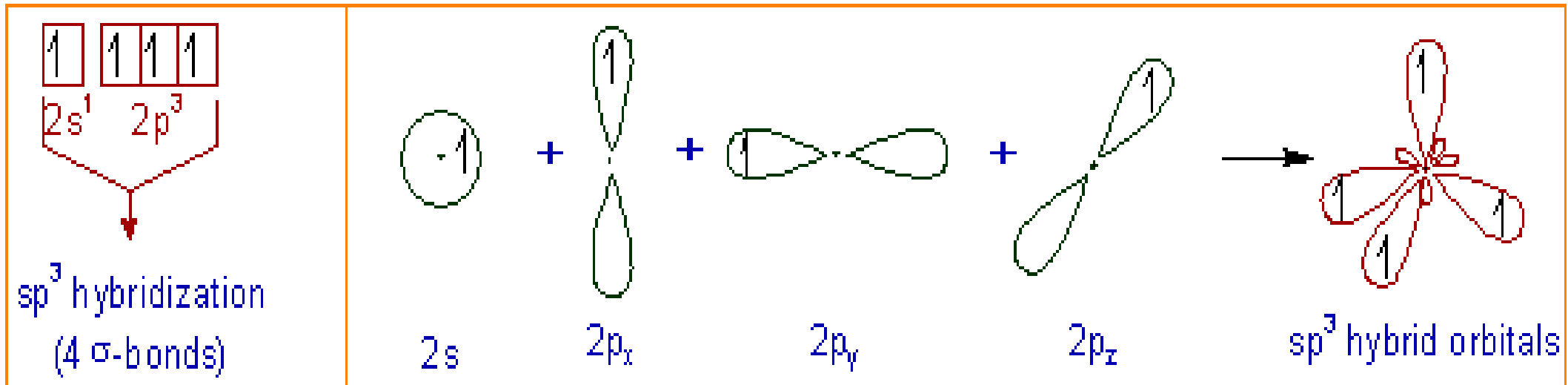
- 1- It is **weaker** than sigma bond.
- 2- Rotation around the bond axis is **restricted**.

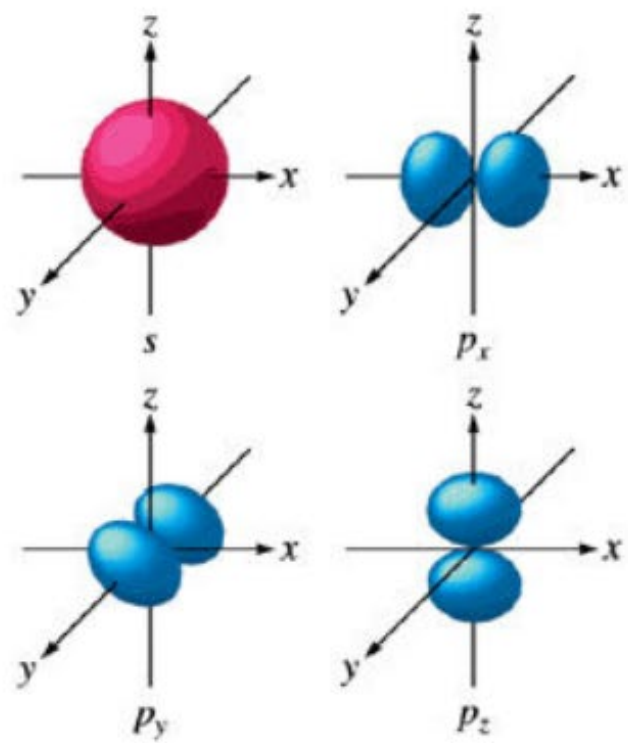
Hybridization

" **Orbitals** in the **same** atom which lie close to one another in energy tend to **combine** or mix and **redistribute** to form equivalent number of new orbitals" the new orbitals are called **hybridized orbitals**.

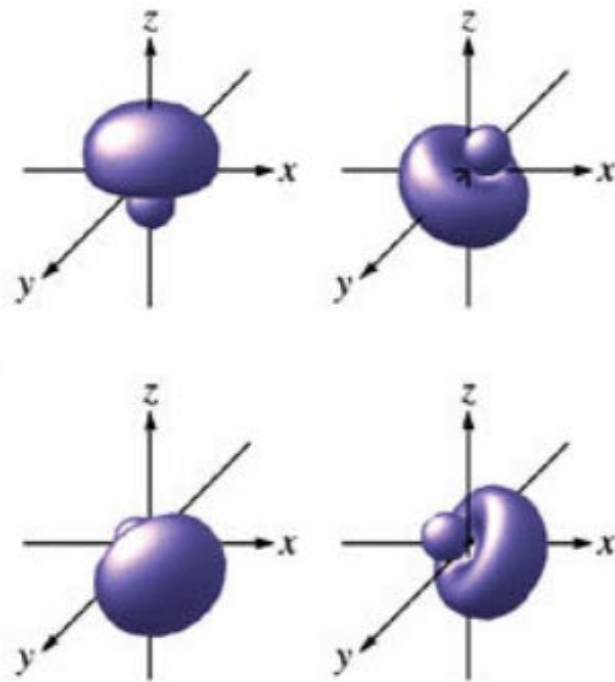
Hybrid Orbitals

- Which are formed by mixing (hybridizing) the carbon's atomic orbitals.
- When carbon atoms bond to 4 other atoms, the 2s orbital and all three 2p orbitals in the valence shell combine to produce four sp^3 hybrid orbitals

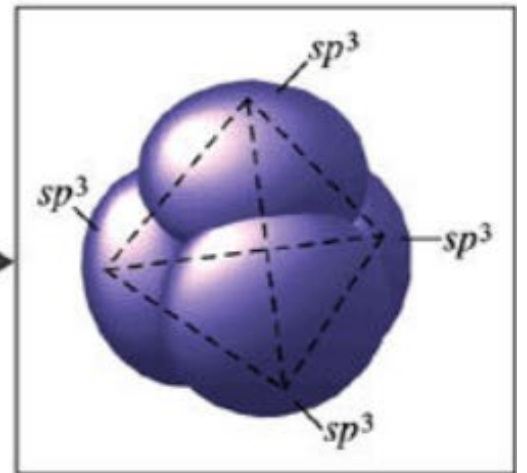




Hybridization

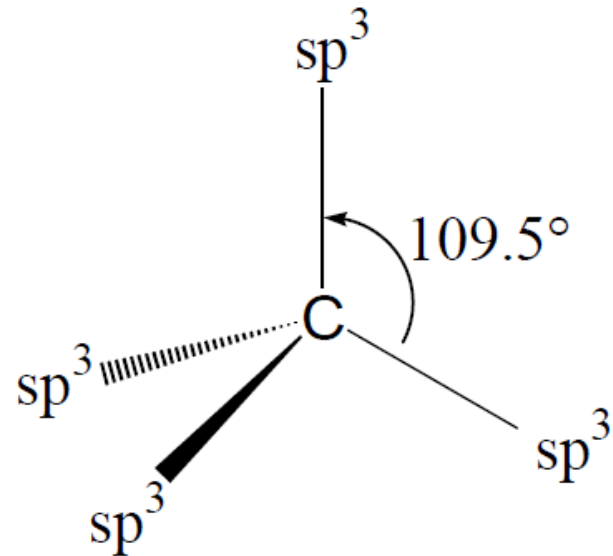
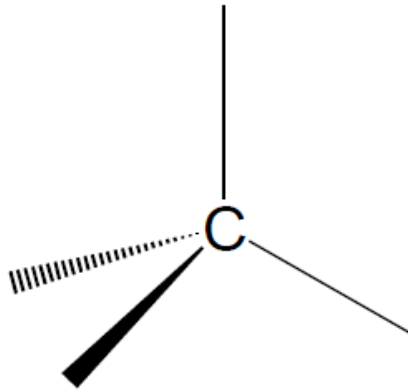
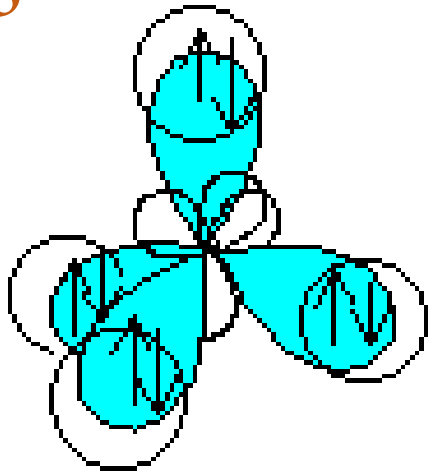


gives a
tetrahedral
arrangement



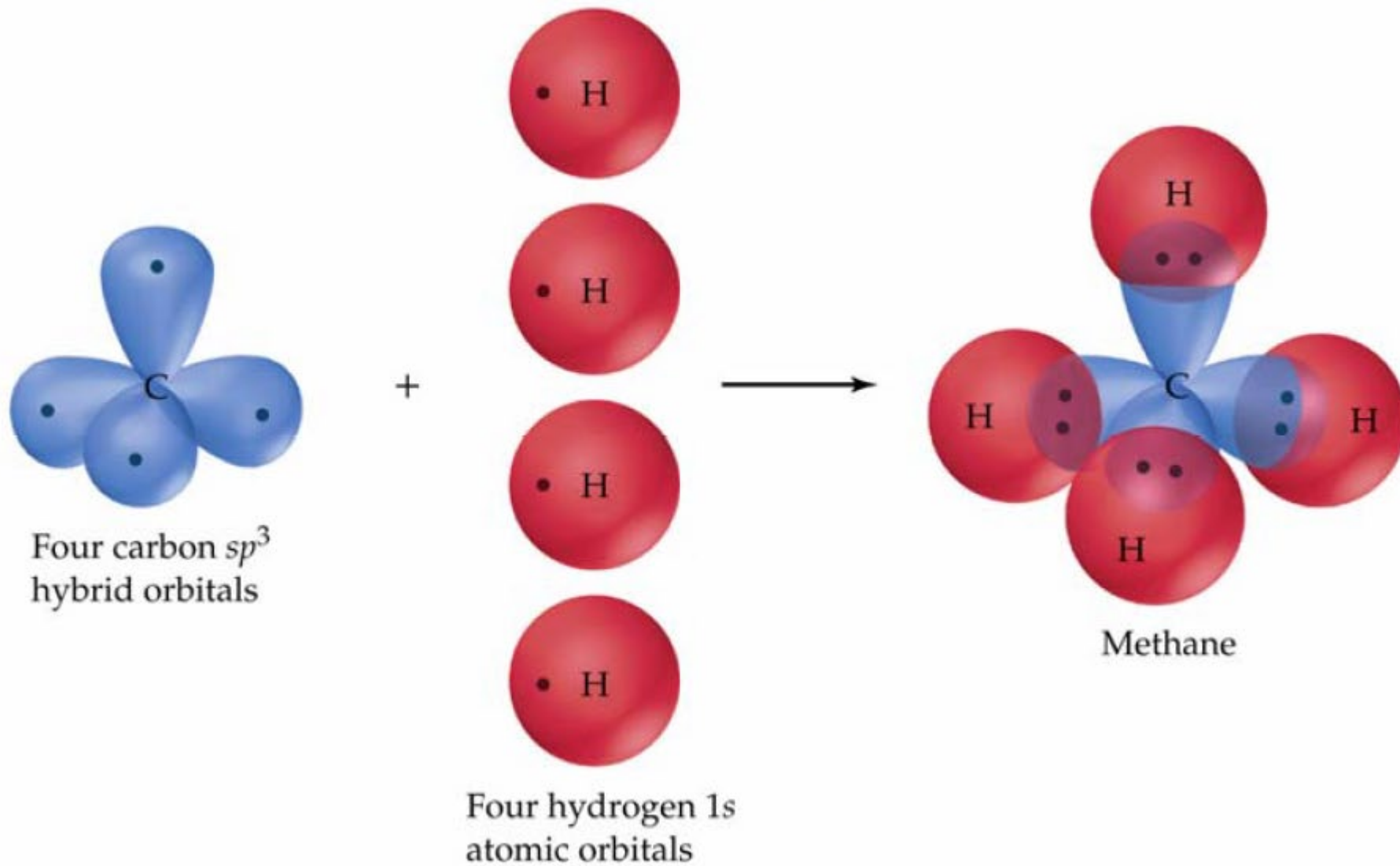
The Shape of an sp^3 Carbon

- In order to get as far away from each other as possible (thus minimizing electron-electron repulsions), the sp^3 orbitals are arranged in the shape of a tetrahedron around the central carbon atom, with bond angles of 109.5°



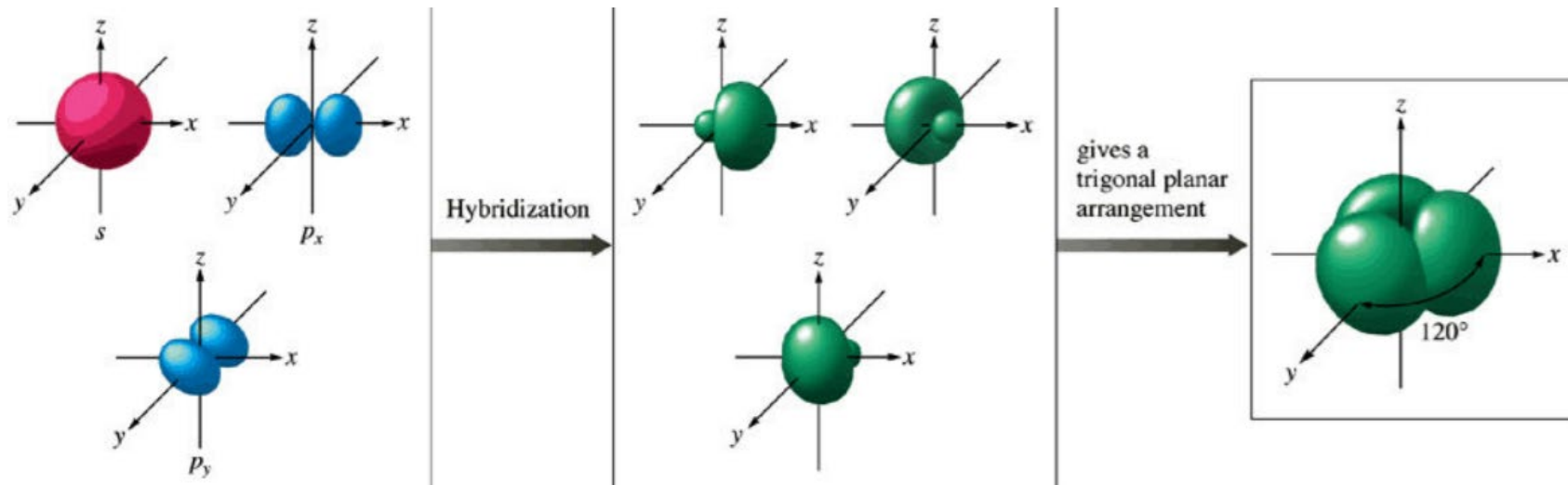
In alkane, SP^3 -hybrid orbitals of the carbon atom form four single bonds to four atoms. Each bond forms an angle of approximately 109.5° with each of the other bonds in the structure. This geometry is a regular tetrahedron .

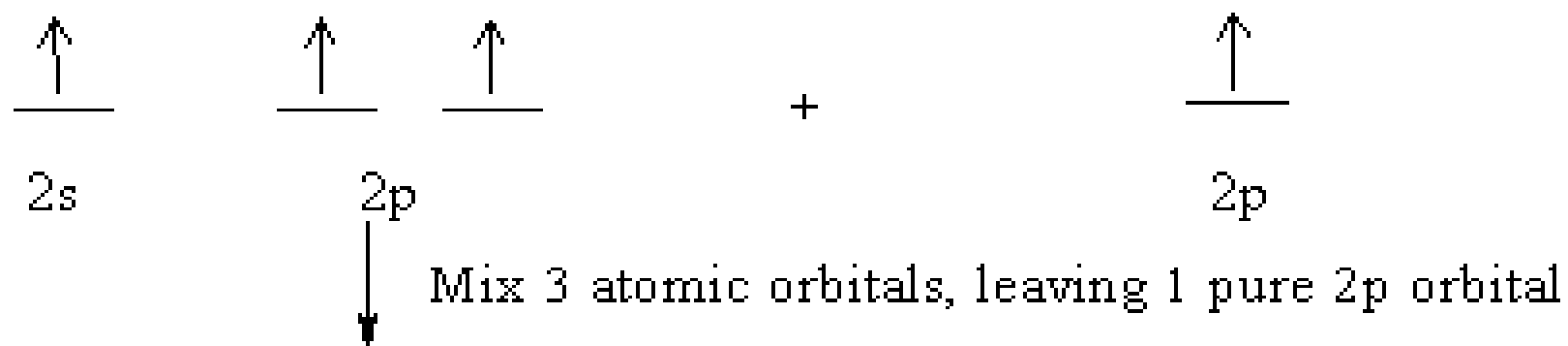
The Shape of an sp^3 Carbon



Sp² hybridized carbon

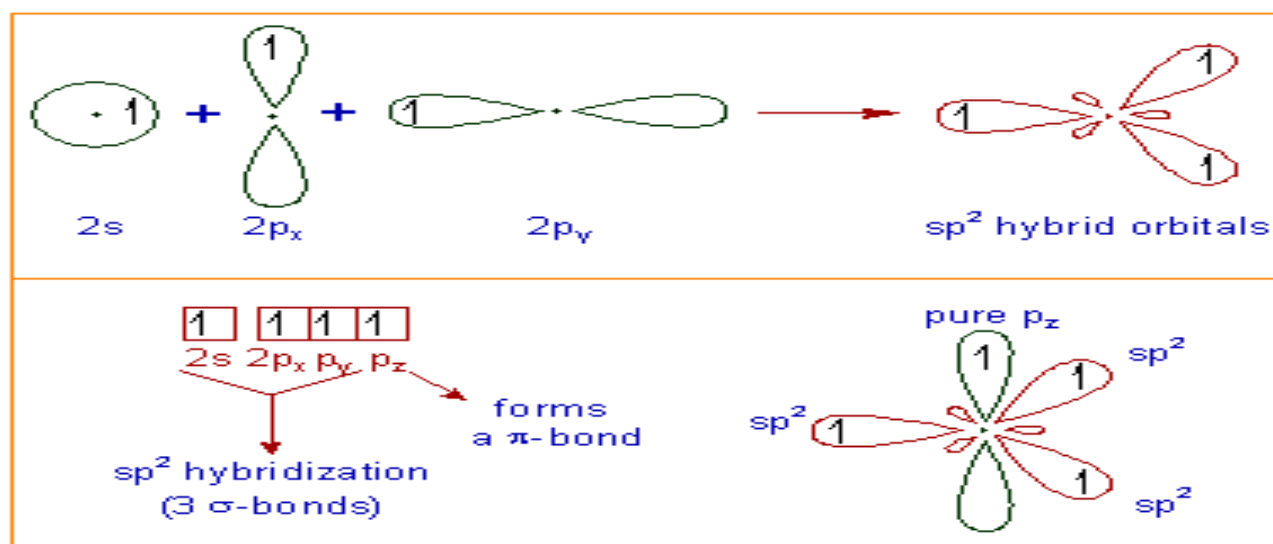
In order to form the 3 sigma bonds in ethene, **one 2s** and **two** of the **2p** orbitals will mix to form **three sp²** hybridized orbitals.



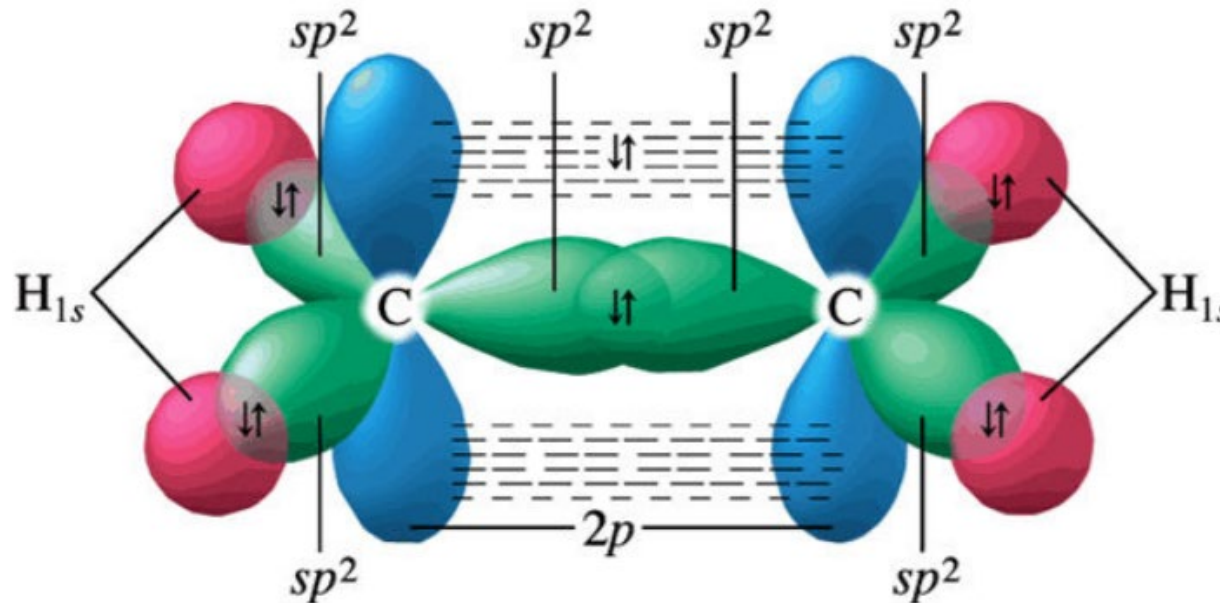


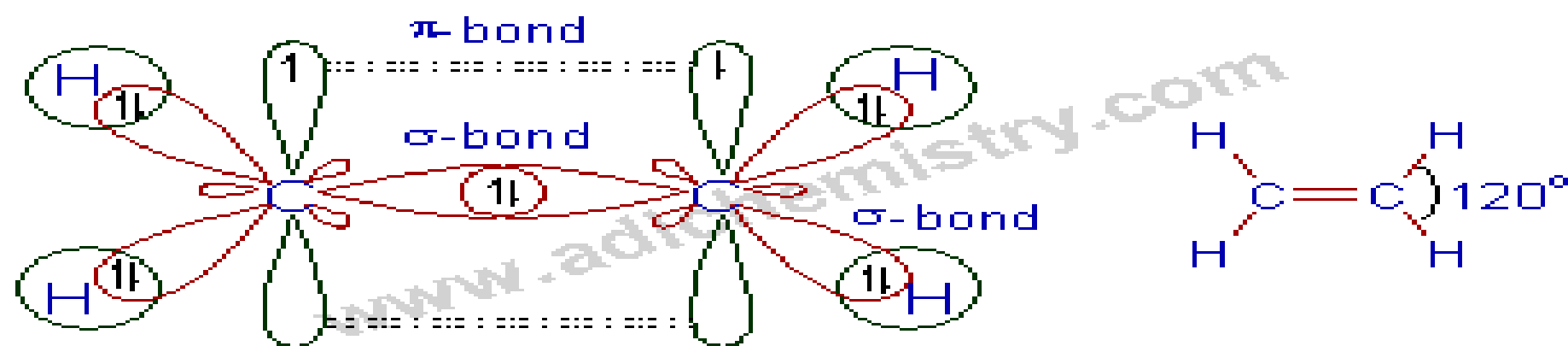
three sp^2 hybrid orbitals have formed, each having one electron

one unmixed pure p orbital having one electron



- The **new sp^2** orbitals are degenerate and in the **same plane** One **2p orbital** is left unchanged and is **perpendicular to that plane**
- sp^2 orbitals are in a plane with 120° angles

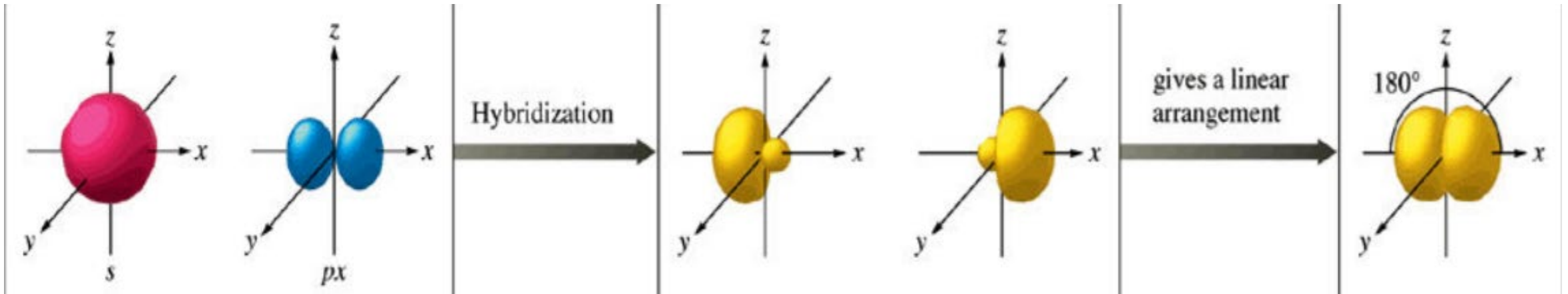




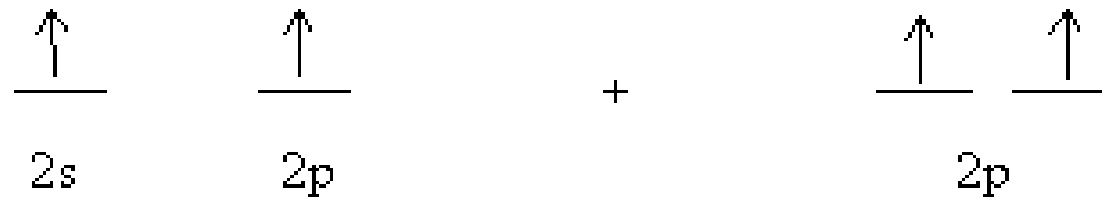
Planar structure of ethylene molecule

sp Hybridization

- **sp Hybridization: combination of an s orbital with an p orbital to form two sp hybrid orbitals with 180° angles**



When a carbon atom is joined to only two other atoms as in acetylene, C₂H₂.



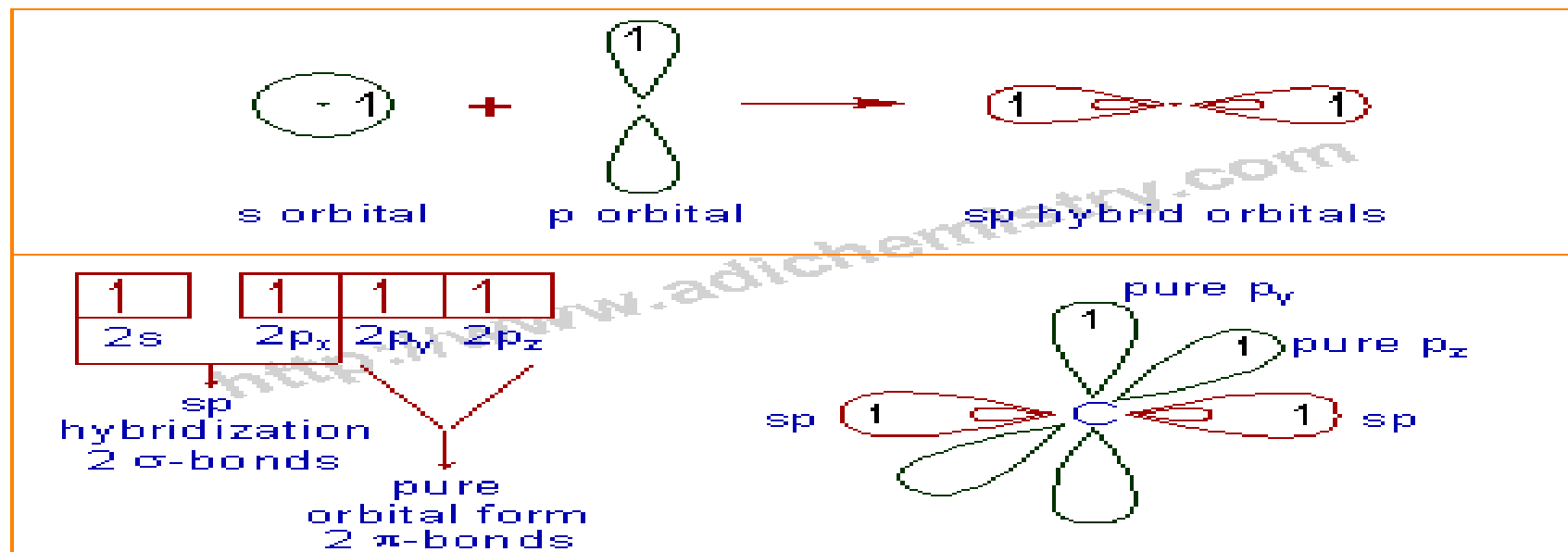
Mix 2 atomic orbitals, leaving 2 pure 2p orbitals



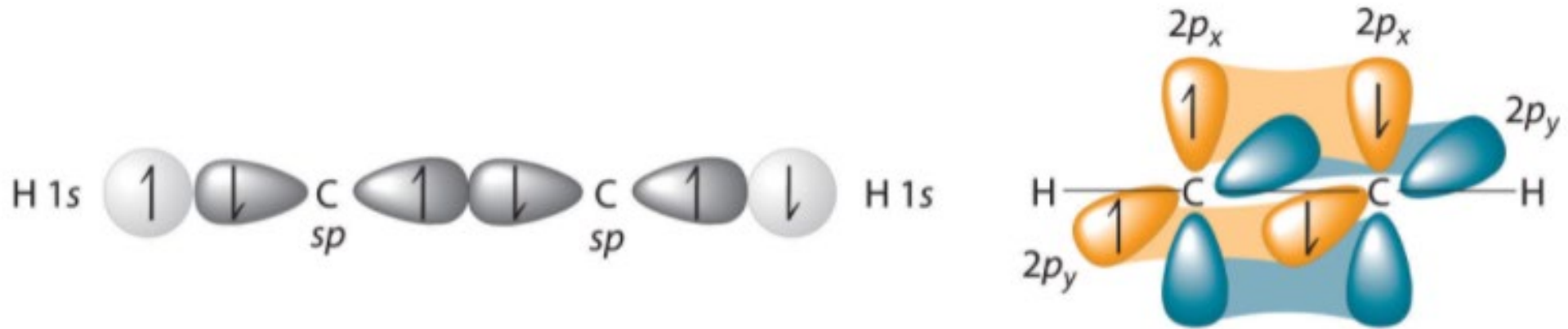
two sp hybrid orbitals have
formed, each having one electron

two unmixed pure p orbitals
each having one electron

The two sp hybrid orbitals arrange themselves in three dimensional spaces to get as far apart as possible. The geometry which achieves is straight line with a **bond angle of 180°** . The two pure p orbitals which were not mixed are perpendicular to each other and to the line of the sp orbitals.



- 2 sp orbitals and two 2p orbitals
- sp orbitals are linear.



Hybridization Types

Hybridization	Number of Hybridized Orbitals	Number of Unhybridized Orbitals	Geometry	Bond Angle
sp^3	4	0	Tetrahedral	109.5°
sp^2	3	1	Trigonal planar	120°
sp	2	2	Linear	180°



Electronegativity

It is a measure of the tendency of an atom of an electronegative element in a molecule to attract electrons to itself.

It depend on:

1- The nuclear charge.

2- Atomic radius.

Increase of nuclear charge and decrease of atomic radius lead to increase of electronegativity.

This is the ability of an atom to attract electrons. *EN* increases in the same period (horizontal row) of the periodic table with the increase in the atomic number (left to right) and decreases as we go down in the vertical column (group).

Electronegativity Trend

1A												0									
1 H 1.00797												2 He 4.0026									
3 Li 6.941	4 Be 9.0122											5 B 10.811	6 C 12.01115	7 N 14.0067	8 O 15.9994	9 F 18.9984	10 Ne 20.179				
11 Na 22.9898	12 Mg 24.305	3B	4B	5B	6B	7B	8B		1B	2B	13 Al 26.9815	14 Si 28.086	15 P 30.9738	16 S 32.064	17 Cl 35.453	18 Ar 39.948					
19 K 39.098	20 Ca 40.08	21 Sc 44.956	22 Ti 47.90	23 V 50.942	24 Cr 51.996	25 Mn 54.9380	26 Fe 55.847	27 Co 58.9332	28 Ni 58.70	29 Cu 63.54	30 Zn 65.38	31 Ga 69.72	32 Ge 72.59	33 As 74.9216	34 Se 78.96	35 Br 79.904	36 Kr 83.80				
37 Rb 85.47	38 Sr 87.62	39 Y 88.905	40 Zr 91.22	41 Nb 92.906	42 Mo 95.94	43 Tc (99)	44 Ru 101.07	45 Rh 102.905	46 Pd 106.4	47 Ag 107.868	48 Cd 112.41	49 In 114.82	50 Sn 118.69	51 Sb 121.75	52 Te 127.60	53 I 126.9045	54 Xe 131.30				
55 Cs 132.905	56 Ba 137.53	57 La 138.91	72 Hf 178.49	73 Ta 180.948	74 W 183.85	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.09	79 Au 196.967	80 Hg 200.59	81 Tl 204.37	82 Pb 207.19	83 Bi 208.980	84 Po (210)	85 At (210)	86 Ra (222)				
87 Fr (223)	88 Ra 226.0254	89 Ac (227)	104 Rf (267)	105 Ha (260)																	

Directions of increasing electronegativity

High Electronegativity

Highly reactive because of their strong tendency to capture electrons.

Periodic Table of the Elements

Highly reactive because they yield electrons easily.

Low Ionization Energy

I	II	Transition Metals										III	IV	V	VI	VII	0
H ¹												B ⁵	C ⁶	N ⁷	O ⁸	F ⁹	Ne ¹⁰
Li ³	Be ⁴											Al ¹³	Si ¹⁴	P ¹⁵	S ¹⁶	Cl ¹⁷	Ar ¹⁸
Na ¹¹	Mg ¹²	III B	IV B	V B	VI B	VII B	VIII B			III	II B	Ga ³¹	Ge ³²	As ³³	Se ³⁴	Br ³⁵	Kr ³⁶
K ¹⁹	Ca ²⁰	Sc ²¹	Ti ²²	V ²³	Cr ²⁴	Mn ²⁵	Fe ²⁶	Co ²⁷	Ni ²⁸	Cu ²⁹	Zn ³⁰	Ga ³¹	Ge ³²	As ³³	Se ³⁴	Br ³⁵	Kr ³⁶
Rb ³⁷	Sr ³⁸	Y ³⁹	Zr ⁴⁰	Nb ⁴¹	Mo ⁴²	Tc ⁴³	Ru ⁴⁴	Rh ⁴⁵	Pd ⁴⁶	Ag ⁴⁷	Cd ⁴⁸	In ⁴⁹	Sn ⁵⁰	Sb ⁵¹	Te ⁵²	I ⁵³	Xe ⁵⁴
Cs ⁵⁵	Ba ⁵⁶		Hf ⁷²	Ta ⁷³	W ⁷⁴	Re ⁷⁵	Os ⁷⁶	Ir ⁷⁷	Pt ⁷⁸	Au ⁷⁹	Hg ⁸⁰	Tl ⁸¹	Pb ⁸²	Bi ⁸³	Po ⁸⁴	At ⁸⁵	Rn ⁸⁶
Fr ⁸⁷	Ra ⁸⁸		Rf ¹⁰⁴	Ha ¹⁰⁵													
Lanthanides		La ⁵⁷ Ce ⁵⁸ Pr ⁵⁹ Nd ⁶⁰ Pm ⁶¹ Sm ⁶² Eu ⁶³ Gd ⁶⁴ Tb ⁶⁵ Dy ⁶⁶ Ho ⁶⁷ Er ⁶⁸ Tm ⁶⁹ Yb ⁷⁰ Lu ⁷¹															
Actinides		Ac ⁸⁹ Th ⁹⁰ Pa ⁹¹ U ⁹² Np ⁹³ Pu ⁹⁴ Am ⁹⁵ Cm ⁹⁶ Bk ⁹⁷ Cf ⁹⁸ Es ⁹⁹ Fm ¹⁰⁰ Md ¹⁰¹ No ¹⁰² Lr ¹⁰³															

Metal
 Metalloid
 Nonmetal

The highly reactive corners of the Periodic Table

Electronegativities of Selected Elements

H 2.2						
Li 1.0	Be 1.6	B 2.0	C 2.6	N 3.0	O 3.4	F 4.0
Na 0.9	Mg 1.3	Al 1.6	Si 1.9	P 2.2	S 2.6	Cl 3.2
K 0.8						Br 3.0
						I 2.7

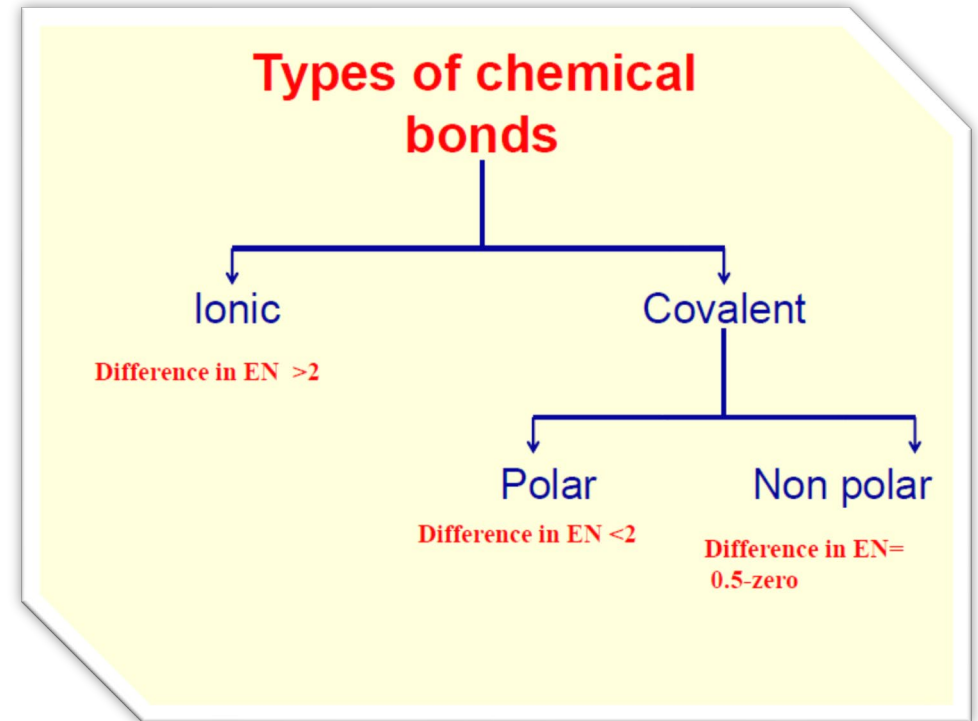
Types of Chemical Bonds

- **Covalent Bonds:**

- Non-polar and polar covalent bonds

- **Ionic Bonds:**

- Transfer of electrons from one atom to another
(e.g., NaCl)



Types of chemical bonds:

1- Ionic bond:

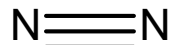
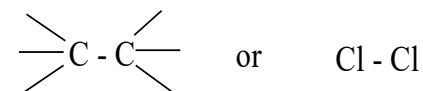
These are usually formed between elements at the extreme of the periodic table, i.e. **formed between atoms widely different in $EN(> 2)$**

e.g. Na^+Cl^- , K^+Br^-

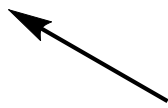
Nonpolar covalent bonds

A **bond** in which the **electrons** are shared **equally**,
When **two identical atoms** (such as two carbon atoms of
a **C-C bond**) **share a pair of electrons**

Typical Molecules with Nonpolar Covalent Bonds:



Electron equally shared

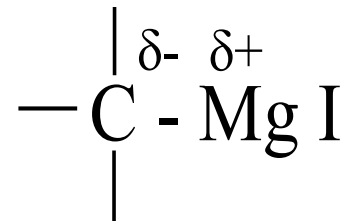
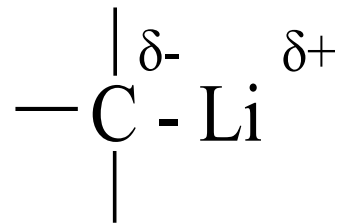
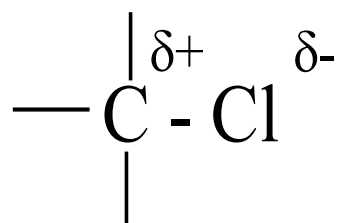
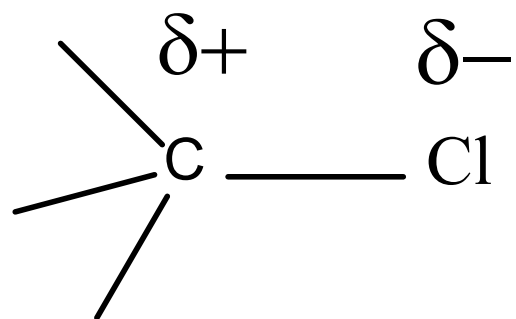


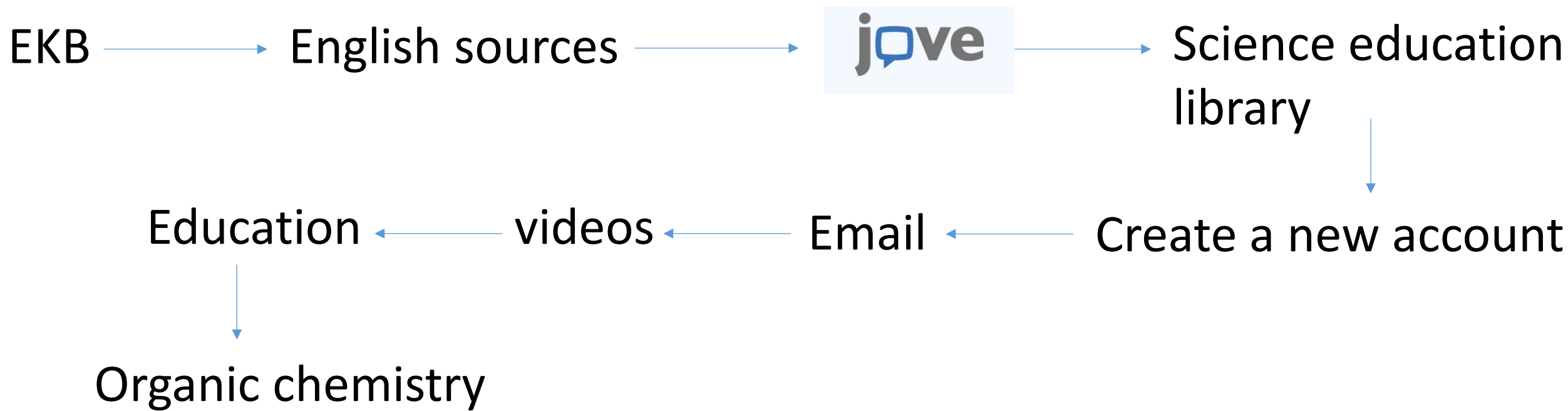
Polar covalent bonds

Covalent bonds between elements that have relatively large differences in electronegativity, the electrons are not shared equally by the two bonded atoms which the shared electrons are pulled closer to one of the bonded atoms.

In polar covalent bond

In covalent bond the more electronegative atom attracts the electrons and carries partially negative charge, and the other atom carries partially positive charge.





Thank you